B.Tech. Batch 2025 Onwards

SEMESTER-1
(COMMON SYLLABUS)

Course Title	**Linear Algebra & Matrix Analysis**	Course Code	**MA01**
Department	**CSE/ECE/DSAI/SCI. & MATHS/HUMANITIES & MGMT.**	Credits, Structure (L-T-P-S)	**3, (2-1-0-0)**
Offered for	**All Programs**	Type	**Compulsory**
Prerequisite	**NA**	Laboratory	No

Course Outcomes

At the end of the course, students should be able to:

1. Solve systems of linear equations using multiple methods, including Gaussian elimination and matrix inversion.
2. Carry out matrix operations, including inverses and determinants.
3. Determine eigenvalues and eigenvectors and solve eigenvalue problems.
4. Apply principles of matrix algebra to linear transformations.
5. Demonstrate understanding of inner products and associated norms.

Contents:

System of linear equations, Inverse and rank of a matrix by using elementary transformations, Properties of determinants

Vector space over real and complex fields, Linear dependence, independence and spanning set, Linear transformation and its matrix, Change of basis, Eigenvalues and eigenvectors. Diagonalization of matrices.

Inner product spaces, Gram-Schmidt orthonormalization, Singular Value Decomposition (SVD).

Text Books

1. Seymour Lipschutz, Marc Lipson, Schaum's outlines of Linear Algebra ,McGraw-Hill Education
2. J. Defranza and D. Gagliardi, Introduction to Linear Algebra with Applications, McGraw-Hill
3. K. Hoffman and R. Kunze, Linear Algebra, Prentice Hall.
4. Strang, Gilbert. Introduction to Linear Algebra. 4th ed. Wellesley-Cambridge Press ,2009.

Course Title	**Calculus**	Course Code	**MA02**
Department	**CSE/ECE/DSAI/SCI. & MATHS/HUMANITIES & MGMT.**	Credits, Structure (L-T-P-S)	**3, (2-1-0-0)**
Offered for	**All Programs**	Type	**Compulsory**
Prerequisite	**NA**	Laboratory	No

Course Outcomes

At the end of the course, students should be able to:

1. Find limits of functions (graphically, numerically and algebraically)
2. Analyse and apply the notions of continuity and differentiability to algebraic and transcendental functions.
3. Determine derivatives using implicit differentiation and use to study characteristics of a curve.
4. Apply the Fundamental theorem of calculus to evaluate definite integrals.
5. Use differentiation and integration to solve real world problems such as rate of change, optimization, and area problems.

Contents:

Review of single variable calculus: Review of Limit, continuity and differentiability of single variable functions, Indeterminate forms and L'Hospital rule, Mean Value theorem, Maclaurin and Taylor series expansions of functions of one variable.

Functions of Several variables: Functions of several variables, Limits and continuity, Partial derivatives and differentiability, Linearization and differentials, Chain rule, Gradient vector, Tangent planes, Directional derivatives, Extreme values and saddle points, Lagrange multipliers, Taylor's formula, Partial derivatives with constrained variables.

Multiple integral: Multiple integral, Double integrals, Change of order of integration, Area and volume by double integral, Double integrals in polar form, Triple integrals in rectangular coordinates, Triple integrals in cylindrical and spherical coordinates, Substitutions in multiple integrals.

Text Books

1. M. D. Weir and J. Hass, Thomas' Calculus, 12 th edition, Pearson.
2. G. B. Thomas and R. L. Finney, Calculus and Analytic Geometry, 9th Ed, Pearson.
3. Huges-Hallett et al, Calculus: Single and Multivariable, 6 th edition, John-Wiley & Sons (USA).
4. J. Stewart, Multivariable Calculus, Hybrid Edition.

Course Title	**Problem Solving with C Programming**	Course Code	**CSE01**
Department	**CSE/ECE/DSAI/SCI. & MATHS/HUMANITIES & MGMT.**	Credits, Structure (L-T-P-S)	**3, (2-0-2-2)**
Offered for	**All Programs**	Type	**Compulsory**
Prerequisite	**NA**	Laboratory	Mandatory 2 Hours

Course Outcomes

At the end of the course, students should be able to:

1. Given a computational problem, identify and list the programming task involved.
2. Implement conditional branching, iteration, and recursion.
3. Use arrays, pointers, and structures to develop algorithms and programs.

Contents:

The basic organization of a computer, Computer Languages, Algorithm, Flowchart, Features of C, Structure of C Program, Basic elements of C programming, Conditional instruction, Iterations, Unconditional statements, Arrays, Stings, Functions, recursion, Pointers, Pointers with arrays, and Functions. Dynamic Memory allocation, Structures and Unions, and Files.

Text Books

1. The C Programming Language Brian W. Kernighan and Dennis M. Ritchie (2nd Edition, 1988), Prentice Hall (ISBN:978-0131103627)
2. Schaum's Outline of Programming with C by Byron S Gottfried (1996), McGraw-Hill Education (ISBN:978-0070240353

Course Title	**Digital Electronics using Verilog**	Course Code	**ECE01**
Department	**CSE/ECE/DSAI/SCI. & MATHS/HUMANITIES & MGMT.**	Credits, Structure (L-T-P-S)	**3, (2-0-2-0)**
Offered for	**All Programs**	Type	**Compulsory**
Prerequisite	**NA**	Laboratory Choice	Mandatory 2 Hours

Course Outcomes

At the end of the course, students should be able to:

1. Understand the basics of any digital systems such as number system, logic gates, Boolean logic simplification, FFs etc.
2. Understand basic digital system design using standard ICs
3. Analyze and design combinational and sequential circuits using above concepts.
4. Perform simple course projects using above design technique

Contents:

Number system and Logic Gates, Truth tables of different logic gates. Boolean Algebra and K-maps, NAND/NOR implementations, Logic Simplification using Tabular method, etc, Combinational Circuit Design: Design procedure, Different Adders and Subtractors-Half Adder, Full Adder, Half Subtractor, Full Subtractor, 4-bit Ripple Carry Adder, Carry Look Ahead Adder, Decoder, Encoders, Multiplexers, De-Multiplexers, Magnitude Comparator, etc. Sequential Circuit Design: Basics, Latches and Flip-flops, conversion from one FF to another, Serial and Parallel shift Registers, Synchronous and Asynchronous Counter Designing, Mealy and Moore Machine.

Text Books

1. R.P. Jain, “Modern Digital Electronics”, 3rd Edition, Tata McGraw Hill, 2003.
2. T.L. Floyd, “Digital Fundamentals”, 10th Edition, Pearson education, 2011
3. M. Morris Mano, Michael D. Ciletti, “Digital Design- with an Introduction to the Verilog HDL”, 5th Ed, 2013, Pearson.

Course Title	**International Language Competence**	Course Code	**HSS01**
Department	**CSE/ECE/DSAI/SCI. & MATHS/HUMANITIES & MGMT.**	Credits, Structure (L-T-P-S)	**2, (1-0-2-2)**
Offered for	**All Programs**	Type	**Compulsory**
Prerequisite	**NA**	Laboratory	Mandatory

Course Outcomes

At the end of the course, students should be able to:

1. Communicate effectively in the foreign language in a variety of speaking situations.
2. Communicate effectively in the foreign language via proficient, articulate, and well-organized writing.
3. Demonstrate comprehension of the spoken foreign language in a variety of listening situations.
4. Demonstrate comprehension of a wide range of foreign language written materials.

Contents:

Overview of the course and evaluation strategies/Importance of Communication skills Introduction to Communication, Understanding the Basics of Communication, Role and importance of Communication Process of Communication and Types of Communication Barriers to communication

Vocabulary Enhancement & Development of Communicative Competence: asking and answering questions, expressing opinions and attitudes narration and description, word formation and idiomatic expressions, Practice Exercises

Reading Exercise - R.K. Laxman's The Gold Frame, G.B.Shaw's Spoken and Broken English, Robert Forst's The Road Not Taken

Introduction to Technical Report Writing: Importance, Guidelines, and Format

Report Writing Practice Session; Discussion and Chapter Review

Text Books

1. R.K. Laxman's The Gold Frame
2. G.B.Shaw's Spoken and Broken English
3. Robert Forst's The Road Not Taken
4. Technical Communication Principles and Practice by Raman and Sharma
5. Class Notes and Practice Sessions

Course Title	**IT Workshop: Full Stack Prototyping with AI Support**	Course Code	**SK01**
Department	**CSE/ECE/DSAI/SCI. & MATHS/HUMANITIES & MGMT.**	Credits, Structure (L-T-P-[C])	**2, (1-0-2-2)**
Offered for	**All Programs**	Type	**Compulsory**
Prerequisite	**NA**	Laboratory Choice	Mandatory

Course Description: This hands-on workshop enables students to design, develop, and deploy end-to-end functional prototypes for web and mobile applications, leveraging AI-powered tools for faster development. The course blends Full Stack development (front-end, back-end, and database) with AI-assisted features such as code generation, UI design, data handling, and intelligent services.

Course Outcomes (COs): On successful completion of this course, students will be able to:

- **CO1:** Apply fundamental concepts of full stack development (frontend, backend, database, APIs) to design functional prototypes.
- **CO2:** Use AI-assisted tools to accelerate coding, interface design, and backend logic development.
- **CO3:** Integrate AI/ML services (e.g., Chatbots, Recommendation Engines, Vision APIs) into full stack prototypes.
- **CO4:** Deploy prototypes to cloud platforms for public access.
- **CO5:** Collaborate effectively in teams, managing workflows using modern DevOps tools and agile principles.

Mapping to Graduate Attributes: Engineering knowledge, Problem Analysis, Design/Development, Modern Tool Usage, Ethics, Communication, and Project Management.

Course Contents:

Module 1 – Introduction & Tools Setup (Week 1) - Full Stack architecture overview (Client–Server–Database), AI-powered development tools: GitHub Copilot, ChatGPT, Cursor, Replit, etc., Setting up environment: Node.js, Python/Flask/Django, React/Vue, VS Code, Git/GitHub

Module 2 – Frontend Development with AI Assistance (Weeks 2–3) - HTML5, CSS3, JavaScript ES6+ basics (AI-assisted code snippets), Component-based UI design using React/Vue, Responsive design frameworks (Bootstrap/Tailwind CSS), AI-assisted prototyping for UI/UX

Module 3 – Backend Development & APIs (Weeks 4–5) - REST API concepts and design, Backend frameworks: Express.js / Flask / Django, AI-assisted code scaffolding for CRUD operations, Authentication & authorization basics

Module 4 – Databases & Integration (Week 6) - SQL (MySQL/PostgreSQL) and NoSQL (MongoDB), AI-assisted query generation and optimization, ORM integration (SQLAlchemy, Mongoose)

Module 5 – AI/ML Service Integration (Weeks 7–8) - Integrating OpenAI, Hugging Face, or Google Cloud AI APIs, Adding chatbots, summarizers, sentiment analysis, or computer vision to prototypes, Data preprocessing using AI tools

Module 6 – Deployment & DevOps Basics (Week 9) - Deploying applications using Vercel, Netlify, or Heroku, Containerization with Docker (optional), AI-assisted CI/CD setup

Module 7 – Project Sprint & Demo Day (Weeks 10–12) - Agile project management tools, (Trello/Jira), Team sprints with AI-assisted task completion, Final demo and peer evaluation

Course Projects (Examples): Students will work in teams to develop **one full stack prototype** with AI-assisted features. Possible project ideas include:

1. **AI-Powered Resume Builder** – Generates resumes using user input, with real-time formatting and download options.
2. **E-Commerce Portal with AI Recommendations** – AI-based product suggestions and chatbot support.
3. **Smart Student Portal** – With AI-assisted assignment summaries and automated FAQs.
4. **Travel Itinerary Planner** – AI-generated itineraries based on preferences, with map integration.
5. **Healthcare Symptom Checker** – Uses NLP models to provide probable condition information.

Text Books:

1. Srinivasa K G, et al., AI-Assisted Programming for Web and Machine Learning, BPB Publishers, 2025.
2. Bates, B., & Downey, C., *Programming with AI Tools: A Developer's Guide*, Manning Publications.

References:

- OpenAI API Documentation – https://platform.openai.com/docs
- Hugging Face API Documentation – https://huggingface.co/docs
- Mozilla Developer Network (MDN) – https://developer.mozilla.org/
- Docker Documentation – https://docs.docker.com/
- Tailwind CSS Docs – https://tailwindcss.com/docs
- GitHub Copilot Guide – https://docs.github.com/copilot

Course Title	**Entrepreneurship and Design Thinking**	Course Code	**MGM01**
Department	**CSE/ECE/DSAI/SCI. & MATHS/HUMANITIES & MGMT.**	Credits, Structure (L-T-P-[C])	**2, (1-0-2-2)**
Offered for	**All Programs**	Type	**Compulsory**
Prerequisite	**NA**	Laboratory Choice	Faculty Choice

Course Outcomes

At the end of the course, students should be able to:

1. Identify the importance of entrepreneurship.
2. To familiarize students with design thinking concepts and principles
3. To ensure students can practice the methods, processes and tools of design thinking.
4. To ensure students can apply the design thinking approach and have ability to model real world situations

Contents:

The Foundations of Entrepreneurship: What's Feeding the Entrepreneurial Fire? The Cultural Diversity of Entrepreneurship. The Power of "Small" Business, The 10 Deadly Mistakes of Entrepreneurship. Putting Failure into Perspective, How to Avoid the Pitfalls. Creativity, Innovation, and Entrepreneurship, Creativity—A Necessity for Survival. Creative Thinking, Barriers to, Creativity How to Enhance Creativity, The Creative Process. Techniques for Improving the Creative Process Protecting Your Ideas.

Introduction to Innovation, Where does Design-Thinking Fit in? The Human Aspects, Thinking Process, Visualization, Journey Mapping, Mind Mapping, Value Chain Analysis, Brainstorming, Customer Co-creation and Learning Launch, Assumption Testing, Rapid Concept Development, Rapid Prototyping, Leveraging Design Thinking for Innovation

Text Books

1. Thomas W. Zimmerer and Norman M. Scarborough: Essentials of Entrepreneurship and Small Business Management (IV Edition), Prentice Hall, 2004.
2. C. Tripathi, P. N. Reddy: Principles of Management Tata McGraw Hill, 4th Edition, 2010.
3. Poornima M Charantimath : Entrepreneurship Development - Small Business Enterprises , 1st Edition, Pearson Education – 2006
4. Roger Martin, "The Design of Business: Why Design Thinking is the Next Competitive Advantage", Harvard Business Press , 2009.
5. Hasso Plattner, Christoph Meinel and Larry Leifer (eds), "Design Thinking: Understand – Improve – Apply", Springer, 2011
6. Idris Mootee, "Design Thinking for Strategic Innovation: What They Can't Teach You at Business or Design School", John Wiley & Sons 2013.

Course Title	**Introduction to Internet of Things**	Course Code	**DSAI02**
Department	**CSE/ECE/DSAI/SCI. & MATHS/HUMANITIES & MGMT.**	Credits, Structure (L-T-P-S)	**2, (1-0-2-2)**
Offered for	**All Programs**	Type	**Compulsory**
Prerequisite	**Sensors & Actuators**	Laboratory Choice	Mandatory

Course Outcomes

At the end of the course, students should be able to:

1. Understand the basic concepts of different measurement & Mechatronics systems, the function, suitability of different Sensors and Actuators and understand in depth the concepts of Input/Output Signal conditioning
2. Understand the Architectural Overview of IoT and Real World Design Constraints
3. Implement interfacing of various sensors with Arduino/Node MCU, Raspberry Pi, etc.
4. Design and Development of prototypes for real-time applications.

Course Objectives:

- Introduce the concepts, architecture, and applications of IoT in an accessible way for first-year students.
- Provide basic skills in IoT prototyping with microcontrollers and sensors.
- Encourage practical experimentation through IoT kits, simulators, and IoT competition portals (e.g., Hackster.io, IoT competitions on Devpost).

Proposed Lecture Plan (14 Hours):

Module	**Topics**	**Hours**
1. Introduction to IoT	IoT definition, history, and evolution; IoT vs traditional embedded systems; Examples from smart cities, healthcare, agriculture, and industry	2
2. IoT Architecture & Components	IoT layered architecture – perception, network, and application layers; Sensors, actuators, microcontrollers, communication modules	2
3. Hardware Platforms	Arduino, ESP32/ESP8266, Raspberry Pi basics; IoT development boards and their features	2
4. Communication Technologies	Wi-Fi, Bluetooth, Zigbee, LoRa, NB-IoT; Choosing the right protocol for an application	2
5. IoT Cloud Platforms	Overview of MQTT protocol, ThingSpeak, Blynk, Google Cloud IoT, AWS IoT Core	2
6. IoT Data & Security	Data collection, basic analytics, visualization; Security challenges and good practices	2
7. Course Wrap-up & Project Launch	Releasing hands-on challenge/problem statement for IoT mini-project; Evaluation process	2

Practical & Challenge-Based Learning: Students will work in teams on **two practical projects/competitions**:

1. **IoT Sensor Data Logging & Visualization**
   - Using Arduino/ESP32 with DHT11 or temperature sensor.
   - Data logging on ThingSpeak or Blynk, visualizing with charts.
2. **IoT Smart Device Control**
   - Simple home automation prototype (LED/relay control via mobile app or web dashboard).
3. **Optional Challenge (Competition)**
   - Participate in Hackster.io / Devpost beginner-level IoT challenges.
   - Example: *Smart plant watering system*, *IoT weather station*, *Campus energy monitoring*.

Tools & Resources:

- **Hardware:** Arduino UNO, ESP32, sensors (DHT11, LDR, PIR), jumper wires, breadboard.
- **Software:** Arduino IDE, MQTT Explorer, Blynk App, ThingSpeak Dashboard.
- **Simulators (for those without hardware):** Tinkercad Circuits, Wokwi.
- **Portals for challenges:** Hackster.io, Devpost, IoT competitions on Kaggle (IoT data analysis).

References:

1. Bahga, A., & Madisetti, V., *Internet of Things: A Hands-On Approach*, Universities Press.
2. Holler, J., Tsiatsis, V., et al., *From Machine-to-Machine to the Internet of Things*, Elsevier.
3. Tinkercad Circuits – https://www.tinkercad.com/circuits
4. Wokwi IoT Simulator – https://wokwi.com/
5. ThingSpeak Documentation – https://thingspeak.com/docs
6. Hackster.io IoT Projects – https://www.hackster.io/projects

Course Title	**Environmental Engineering & SDG**	Course Code	**VAC02**
Department	**CSE/ECE/DSAI/SCI. & MATHS/HUMANITIES & MGMT.**	Credits, Structure (L-T-P-S)	**(1-0-0-0)**
Offered for	**All Programs**	Type	**Compulsory**
Prerequisite	**NA**	Laboratory Choice	Faculty Choice

Course Outcomes

At the end of the course, students should be able to:

1. Understand the concepts and methods from ecological and physical sciences and their application in environmental problem solving.
2. Understand the transnational character of environmental problems and ways of addressing them, including interactions across local to global scales.
3. Articulate the interdisciplinary context of environmental issues.
4. Identify and justify key stakeholders in humanities and social sciences that need to be a part of sustainable solutions.
5. Formulate an action plan for sustainable alternatives that integrate science, humanist, and social perspectives.

Contents:

Earth Systems:- Earth Science Concepts, Environment, Lithosphere, Hydrosphere, Atmosphere, Biosphere and their interactions with each other, Soil and Soil Dynamics – Soil formation, structure, properties

Natural Resources:- Renewable and non-renewable resources, Major natural resources – forest, water, mineral, Food, energy and land resources

Ecosystem - The Living World :- Ecosystem concept, Nature, Structure, Food chain, Food web, Trophic structure, Energy Flow, Biogeochemical Cycles, Major ecosystems – forest, grassland, aquatic, deserts

Biodiversity :- Biodiversity, factors determining degree of diversity, types, importance, threats to biodiversity, conservation of biodiversity

Environmental Pollution :- Pollution Types: Air pollution, Noise pollution, Water pollution, Soil pollution, Nuclear Pollution Impacts on the Environment and Human Health, Economic Impacts,

Solid waste management, Role of an individual in preventing pollution

Human Population and the Environment :- Population growth: an overview, Population characteristics and variations among nations, Demographic transition, Population explosion, Theories on population growth, Population stabilization, Environment and human health

Social Issues and SDG:- Sustainable and unsustainable development, Measures of sustainable development, Urban problems related to energy, Consumerism, Water conservation, Rain water harvesting, Watershed management, Rehabilitation and resettlement, Climate Change, Global warming, Acid rain, Ozone layer depletion, Environmental ethics, Environmental legislation

Green Computing:- Meaning and benefits of green computing, Approaches to Green Computing – Green manufacturing, Green Design, Green usage practices, Green Disposal, Green Applications **Text Books**

1. Anubha Kaushik & C. P. Kaushik, Perspectives in Environmental Studies, New Age International Publishers
2. Gilbert M. Masters, Introduction to Environmental Engineering & Science, Pearson Education
3. P. K. Sinha & Priti Sinha, Information Technology: Theory and Practice, PHI
4. Bharucha Erach for UGC, Environmental Studies

SEMESTER-2

(COMMON SYLLABUS)

Course Title	**Quantum Mechanics**	Course Code	**SC01**
Department	**CSE/ECE/DSAI/SCI. & MATHS/HUMANITIES & MGMT.**	Credits, Structure (L-T-P-S)	**2, (2-1-0-0)**
Offered for	**All Programs**	Type	**Compulsory**
Prerequisite	**NA**	Laboratory Choice	Faculty Choice

Course Outcomes

At the end of the course, students should be able to:

1. Use with the language of basic quantum mechanics and formalism.
2. carry out calculations of atoms and molecules and describe quantum phenomenon within the electron physics from quantum mechanical relations.
3. carry out spectroscopic studies of different subjects and interpret the results in quantized units.
4. report on applications of quantum physics in nature, technical developments and society.

Contents:

Basics of Quantum Mechanics: Basic principles of quantum mechanics, wave function and the uncertainty principle, probability wave amplitude, probability density, wave equation and Schrodinger formalism, time-independent and time-dependent Schrodinger equations, eigenstate and eigenvalues, expectation values.

Wave Formalism of Quantum Mechanics: Schrodinger Equation in one dimension, and applications. Hydrogen Atom (Spherically symmetric potential) : Orbital and spin angular momentum operators, eigenstates and eigenvalues of angular momentum, central potential, solutions of Schrodinger equation in a central potential and applications (Hydrogen-like atom).

Dirac's bra and ket algebra: Introduction, bra and ket notation, angular momenta, Clebsch-Gordon coefficients, spin-orbit interaction and applications. Hilbert space and concept of Qubits.

Text Book:

1. Ajoy Ghatak and S. Lokanathan; Quantum Mechanics: Theory and Application

Reference Book:

1. Griffiths, D.J.; Introduction to Quantum Mechanics, 2nd edition: Pearson Intern.
2. Leonard I. Schiff; Quantum Mechanics

Course Title	**Probability & Statistics**	Course Code	**MA03**
Department	**CSE/ECE/DSAI/SCI. & MATHS/HUMANITIES & MGMT.**	Credits, Structure (L-T-P-S)	**3, (2-2-0-0)**
Offered for	**All Programs**	Type	**Compulsory**
Prerequisite	**NA**	Laboratory Choice	Faculty Choice

Course Outcomes

At the end of the course, students should be able to:

1. Use the basic probability rules, including additive and multiplicative laws, using the terms, independent and mutually exclusive events.
2. Analyze statistical data using measures of central tendency, dispersion and location.
3. Identify the characteristics of different discrete and continuous distributions.
4. Translate real-world problems into probability models.
5. Identify the type of statistical situation to which different distributions can be applied.

Contents:

Axiomatic definition of probability, Probability on finite sample spaces, Joint and conditional probabilities, independence, total probability; Bayes' rule and applications.

Random variables, Definition of random variables, continuous and discrete random variables, Bivariate and Multivariate random variables, Sampling and Sampling Distributions, Central Limit Theorem;

Cumulative distribution function (cdf) for discrete and continuous random variables; probability mass function (pmf); Probability density functions (pdf) and properties, Jointly distributed random variables, conditional and joint density and distribution functions, independence.

Measures of central tendency, Dispersion, Skewness, Kurtosis, Data Representation using Histogram, Pie Chart, Boxplot, Biplot, Multidimensional Scaling etc.

Methods of Estimation, Unbiased estimators; Confidence Interval Estimation: Z-interval, t-interval; Hypothesis Testing, Types of Errors, Rejection Region Approach and p-value Approach.

Text Books

1. Anthony J Hayter; Probability and Statistics for Engineers, Cengage learning, 4th Edition, 2012.
2. S. C. Gupta and V. K. Kapoor, Fundamentals of Mathematical Statistics, S. Chand and Sons, 11th Edition, 2014.
3. Paul L. Meyer, Addison-Wesley, Introductory Probability and Statistical Applications, 1966.
4. Sheldon M. Ross, Introduction to Probability and Statistics for Engineers and Scientists, 5th Edition, Academic Press, 2014.

Course Title	**Data Structures**	Course Code	**CSE02**
Department	**CSE/ECE/DSAI/SCI. & MATHS/HUMANITIES & MGMT.**	Credits, Structure (L-T-P-S)	**3, (2-0-2-2)**
Offered for	**CSE / DSAI / ECE**	Type	**Compulsory**
Prerequisite	**Introduction to Programming**	Laboratory Choice	Mandatory

Course Outcomes

At the end of the course, students should be able to:

1. Illustrate arrays and structures with programming solutions for real time problems
2. Quote the implication of stacks and queues for different problems
3. Propose programming solutions using variations of stacks and queues
4. Manage data set operations using variations of linked list
5. Devise application to solve tree-oriented problems and develop solutions for problems based on graphs

Contents:

Basic Concepts: Stacks, Stacks Using Dynamic Arrays, Queues, Circular Queues. Evaluation of Expressions. Linked Lists: Singly Linked lists, Doubly Linked lists, Circular linked lists. Polynomials, Additional List operations. Trees: Introduction, Binary Trees, Binary Tree Traversals, additional Binary Tree Operations, Heaps, Binary Search Trees. Graphs: The Graph Abstract Data Type, Elementary Graph Operations. Searching and Sorting. Hashing and its implementations.

Text Books

1. Horowitz, Sahni, Anderson-Freed: Fundamentals of Data Structures in C, 2nd Edition, Universities Press, 2008.
2. Yedidyah, Augenstein, Tannenbaum: Data Structures Using C and C++, 2nd Edition, Pearson Education, 2003.
3. Data Structures, Seynour Lipschutz and GAV Pai, Schaum's Outlines, McGraw Hill, 2008.
4. Richard F. Gilberg and Behrouz A. Forouzan: Data Structures A Pseudocode Approach with C, Cengage Learning, 2005

Course Title	**Signals and Systems**	Course Code	**ECE02**
Department	**CSE/ECE/DSAI/SCI. & MATHS/HUMANITIES & MGMT.**	Credits, Structure (L-T-P-S)	**3, (2-0-2-0)**
Offered for	**All Programs**	Type	**Compulsory**
Prerequisite	**NA**	Laboratory Choice	Faculty Choice

Course Outcomes

At the end of the course, students should be able to:

1. Understand different types of signals-continuous and discrete, odd and even, periodic and aperiodic etc.
2. Classify systems based on their properties
3. Analyse continuous time signals and systems by using appropriate mathematical tools
4. Analyse sampling process and sampling of discrete time signals.
5. Determine Fourier transforms for continuous-time and discrete-time signals

Contents:

Signals: classification of signals; signal operations: scaling, shifting and inversion; signal properties; elementary signals.

Systems: classification of systems; system properties; continuous-time linear time invariant (LTI) and discrete-time linear shift invariant (LSI) systems: impulse response and step response; response to an arbitrary input: convolution; system representation using differential and difference equations; Eigen functions of LTI systems, frequency response and its relation to the impulse response.

Fourier series representation of continuous-time and discrete-time signals, continuous-time Fourier transform and its properties; Parseval's relation, time-bandwidth product; discrete-time Fourier transform and its properties; relations among various Fourier representations, Time and Frequency characterization of Signals and Systems, DFT.

Sampling theorem; aliasing; signal reconstruction: ideal interpolator, zero-order hold, first-order hold.

Definition, region of convergence, properties; transform-domain analysis of LTI/LSI systems, system function: poles and zeros; stability.

Text Books

1. A.V. Oppenheim, A.S. Willsky and H.S. Nawab, "Signals and Systems", Prentice Hall of India, 2006
2. B. P. Lathi,"Signal Processing and Linear Systems", Oxford University Press, 1998.
3. M. J. Roberts, "Fundamentals of Signals and Systems", Tata McGraw Hill, 2007.
4. R.F. Ziemer, W.H. Tranter and D.R. Fannin, "Signals and Systems - Continuous and Discrete", 4/e, Prentice Hall, 1998.

Course Title	**Introduction to AI & ML**	Course Code	**DSAI01**
Department	**CSE/ECE/DSAI/SCI. & MATHS/HUMANITIES & MGMT.**	Credits, Structure (L-T-P-S)	**2, (1-0-2-2)**
Offered for	**All Programs**	Type	**Compulsory**
Prerequisite	**Linear Algebra and Probability**	Laboratory Choice	Mandatory

Course Outcomes

At the end of the course, students should be able to:

1. Understand how AI is different from machine learning
2. Appraise the underlying mathematical relationships within machine learning algorithms and the paradigms of supervised and unsupervised learning
3. Decide what kind of machine learning paradigm can be applied to solve the underlying real-world application
4. Distinguish a game situation from pure individual decision problem

Course Objectives:

- Introduce the fundamental concepts and applications of AI and ML in an intuitive, non-mathematical manner.
- Familiarize students with AI/ML workflows and popular tools through real-world datasets and competitions.
- Encourage curiosity and self-learning through open platforms like Kaggle, AIcrowd, and Google Colab.

Proposed Lecture Plan (14 Hours):

Module	Topics	Hours
1. Introduction to AI & ML	History & evolution of AI; Difference between AI, ML, and DL; Real-world applications in different sectors	2
2. Data and Problem Formulation	Understanding datasets, features, labels; Types of ML problems – classification, regression, clustering	2
3. ML Workflow Basics	Data preprocessing, train/test split, model training, evaluation metrics (accuracy, precision, recall, F1)	2
4. Core Algorithms (Intuitive)	Linear regression, decision trees, k-NN, Naïve Bayes; Intro to neural networks	3
5. AI Tools & Platforms	Google Colab, Jupyter, Kaggle kernels & competitions, AIcrowd	2
6. Ethics & Responsible AI	Bias in data, fairness, explainability, data privacy	1
7. Course Wrap-up & Competition Launch	Problem statement release for class competition, expectations, evaluation	2

Practical & Competition-Based Learning: Students will work on at least **two mini-projects/competitions** during lab/practical sessions:

1. **Beginner-level Kaggle Competition**
   - Example: *Titanic – Machine Learning from Disaster* (binary classification)
   - Goal: Submit predictions, view leaderboard, learn iterative improvement.
2. **Custom AI Challenge**
   - Could be hosted internally or via AIcrowd/DrivenData.
   - Examples: Handwritten digit recognition (MNIST), basic sentiment analysis, house price prediction.
3. **Open Exploration**
   - Teams can optionally pick their own small dataset and apply ML pipeline steps.

Tools & Resources:

- **Platforms:** Kaggle, AIcrowd, DrivenData, Google Colab
- **Libraries:** scikit-learn, pandas, matplotlib, seaborn, TensorFlow/PyTorch (basic exposure only)
- **Datasets:** From competition portals or open datasets (UCI ML Repository, Kaggle Datasets)

References:

1. Géron, A., *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, O'Reilly. *(select chapters for intuition)*
2. Raschka, S., *Python Machine Learning*, Packt Publishing. *(for self-learners)*
3. Kaggle Learn Micro-Courses – https://www.kaggle.com/learn
4. AIcrowd Challenges – https://www.aicrowd.com/challenges
5. Google Colab – https://colab.research.google.com/

Course Title	**Communication and Discourse Strategies**	Course Code	**HSS02**
Department	**CSE/ECE/DSAI/SCI. & MATHS/HUMANITIES & MGMT.**	Structure (L-T-P-[C])	**(1-0-2-[2])**
Offered for	**All Programs**	Type	**Compulsory**
Prerequisite	**IT Workshop**	Laboratory Choice	Mandatory

Course Outcomes

At the end of the course, students should be able to:

1. Identify Common Errors and Rectify Them
2. Develop and Expand Writing Skills through Controlled and Guided Activities
3. Develop Coherence, Cohesion and Competence in Oral Discourse through Intelligible Pronunciation.
4. Ability to handle the interview process confidently
5. Conduct an effective group discussion

Contents:

Overview of the course and evaluation strategies, Introduction to Communication Processes and Models, Forms of Communication; Communication Levels, Ways of Establishing Credible Communication, Evolving Discourse Strategies and Listening Skills

Domain Vocabularies: Building special Vocabularies relating to issues of Commerce/Finance/Politics/Current Affairs/Law/Science & Technology, Common Errors: Indianism, Building advanced vocabulary

Communication Strategies in Contemporary Society: Advertising as Persuasive Process Argument in Science; Argument in Literature, Argument on Stage and in Film

Discourse Markers: Group Discussions; Effective presentations, Dialogues and Debates etc. Contemporary Functional English: Letters and Written Communications; Précis Writing

Text Books

1. Gumperz, John J. Discourse strategies. No. 1. Cambridge University Press, 1982.
2. Scollon, Ron, Suzanne Wong Scollon, and Rodney H. Jones. Intercultural communication: A discourse approach. John Wiley & Sons, 2012.

Course Title	**Human Values and Ethics**	Course Code	**VAC01**
Department	**CSE/ECE/DSAI/SCI. & MATHS/HUMANITIES & MGMT.**	Credits, Structure (L-T-P-[C])	**2, (1-0-2-2)**
Offered for	**All Programs**	Type	**Compulsory**
Prerequisite	**NA**	Laboratory Choice	Faculty Choice

Course Outcomes

At the end of the course, students should be able to:

1. Understand the significance of value inputs in a classroom and start applying them in their life and profession
2. Distinguish between values and skills, happiness and accumulation of physical facilities, the Self and the Body, Intention and Competence of an individual, etc.
3. Understand the value of harmonious relationship based on trust and respect in their life and profession
4. Understand the role of a human being in ensuring harmony in society and nature.
5. Distinguish between ethical and unethical practices, and start working out the strategy to actualize a harmonious environment wherever they work.

Contents:

Value Education, Definition, Concept and Need for Value Understanding the need, basic guidelines, content and process for Value Education; Self Exploration–what is it? - its content and process; 'Natural Acceptance' and Experiential Validation- as the mechanism for self-exploration; Continuous Happiness and Prosperity- A look at basic Human Aspirations; Right understanding, Relationship and Physical Facilities- the basic requirements for fulfillment of aspirations of every human being with their correct priority; Understanding Happiness and Prosperity correctly- A critical appraisal of the current scenario; Method to fulfill the above human aspirations: understanding and living in harmony at various levels

Understanding human being as a co-existence of the sentient 'I' and the material 'Body'; Understanding the needs of Self ('I') and 'Body' - Sukh and Suvidha; Understanding the Body as an instrument of 'I' (I being the doer, seer and enjoyer); Understanding the characteristics and activities of 'I' and harmony in 'I'; Understanding the harmony of I with the Body: Sanyam and Swasthya; correct appraisal of Physical needs, meaning of Prosperity in detail; Programs to ensure Sanyam and Swasthya

Understanding harmony in the Family- the basic unit of human interaction; Understanding values in human-human relationship; meaning of Nyaya and program for its fulfillment to ensure Ubhay-tripti; Trust (Vishwas) and Respect (Samman) as the foundational values of relationship; Understanding the meaning of Vishwas; Difference between intention and competence; Understanding the meaning of Samman, Difference between respect and differentiation; the other salient values in relationship; Understanding the harmony in the society (society being an extension of family): Samadhan, Samridhi, Abhay, Sah-astitva as comprehensive Human Goals; Visualizing a universal harmonious order in society- Undivided Society (AkhandSamaj), Universal Order (SarvabhaumVyawastha)- from family to world family!

Understanding the harmony in the Nature; Interconnectedness and mutual fulfillment among the four orders of nature- recyclability and self-regulation in nature; Understanding Existence as Co-existence (Sah-astitva) of mutually interacting units in all-pervasive space; Holistic perception of harmony at all levels of existence

Natural acceptance of human values; Definitiveness of Ethical Human Conduct; Basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order; Competence in Professional Ethics; Case studies of typical holistic technologies, management models and production systems; Strategy for transition from the present state to Universal Human Order

Text Books

1. R R Gaur, R Sangal, G P Bagaria, 2009, A Foundation Course in Human Values and Professional Ethics.
2. P L Dhar, RR Gaur, 1990, Science and Humanism, Commonwealth Publishers.
3. A N Tripathy, 2003, Human Values, New Age International Publishers.
4. E G Seebauer & Robert L. Berry, 2000, Fundamentals of Ethics for Scientists & Engineers, Oxford University Press

Course Title	**3D Printing and Prototype Design**	Course Code	**TBD**
Department	**CSE/ECE/DSAI/SCI. & MATHS/HUMANITIES & MGMT.**	Credits, Structure (L-T-P-S)	**(0-0-2-2)**
Offered for	**All Programs**	Type	**Compulsory**
Prerequisite	**IT Workshop**	Laboratory Choice	Mandatory

Course Objectives:

- Introduce the fundamentals of additive manufacturing and 3D printing technologies.
- Familiarize students with design workflows for rapid prototyping.
- Provide hands-on experience in modeling, slicing, and printing functional parts.
- Enable students to evaluate prototypes for functional and aesthetic performance.

Course Outcomes (COs):

On successful completion, students will be able to:

- **CO1:** Explain the principles, processes, and applications of 3D printing.
- **CO2:** Design 3D models suitable for additive manufacturing.
- **CO3:** Operate 3D printers and slicing software to produce prototypes.
- **CO4:** Evaluate printed prototypes for quality and usability.
- **CO5:** Apply 3D printing in problem-solving and product development.

Theory Plan – 14 Hours

Module	Topics	Hours
1. Introduction to Additive Manufacturing	History, evolution, and advantages over traditional manufacturing; Key applications in engineering, medicine, art	2
2. 3D Printing Technologies	FDM, SLA, SLS, DMLS; Materials (PLA, ABS, PETG, resins, metals); Choosing the right technology	2
3. Design Principles for 3D Printing	Design for Additive Manufacturing (DfAM); CAD modeling considerations (overhangs, wall thickness, tolerances)	2
4. Slicing & Printer Workflow	Slicing software (Cura, PrusaSlicer); Print settings (layer height, infill, supports); File formats (STL, OBJ, G-code)	2
5. Post-Processing Techniques	Support removal, sanding, priming, painting, vapor smoothing	1
6. Quality Control & Troubleshooting	Common printing defects and fixes; Dimensional accuracy checks	2
7. Applications & Future Trends	Rapid prototyping, mass customization, bioprinting, construction-scale printing	2
8. Ethics, Sustainability & IP	Environmental considerations, material recycling, design copyright	1

Practical Plan – 14 Hours

Practical	Activity	Hours
P1: Introduction to 3D Printer Hardware	Understanding printer components, safety protocols, maintenance basics	2
P2: CAD Modeling Basics	Creating simple parts using TinkerCAD/Fusion 360/FreeCAD	2
P3: Design for Printability	Modifying models for print optimization (supports, infill, scaling)	2
P4: Slicing Practice	Importing models, setting print parameters in Cura or PrusaSlicer	2
P5: Printing First Prototype	Loading filament, starting print, monitoring process	2
P6: Post-Processing	Removing supports, sanding, painting, assembly	2
P7: Prototype Evaluation	Dimensional checks, functionality tests, design improvement discussion	2

Mini Project:

Teams design and print a functional prototype solving a small problem (e.g., phone stand, desk organizer, drone frame, gear assembly).

Evaluation based on **design complexity, print quality, and functional performance**.

Text Books:

1. Gibson, I., Rosen, D.W., & Stucker, B., *Additive Manufacturing Technologies*, Springer.
2. Chua, C.K., & Leong, K.F., *3D Printing and Additive Manufacturing: Principles and Applications*, World Scientific.

References:

- 3D Printing Handbook – 3DHubs
- Autodesk Fusion 360 Tutorials – https://www.autodesk.com/
- Ultimaker Cura Documentation – https://support.ultimaker.com/
- Prusa Knowledge Base – https://help.prusa3d.com/

Course Title	**Constitution of India**	Course Code	**TBD**
Department	**CSE/ECE/DSAI/SCI. & MATHS/HUMANITIES & MGMT.**	Credits, Structure (L-T-P-S)	**(1-0-0-0)**
Offered for	**All Programs**	Type	**Compulsory**
Prerequisite	**-**	Laboratory Choice	Mandatory

Course Objectives:

- Provide an overview of the origin, structure, and key features of the Indian Constitution.
- Create awareness of citizens' rights, duties, and the role of governance.
- Enable students to connect constitutional principles with technology, ethics, and society.

Course Outcomes (COs): By the end of this course, students will be able to:
CO1: Summarize the historical background and philosophy of the Constitution.
CO2: Describe fundamental rights, duties, and directive principles.
CO3: Identify the structure and functioning of the Union, State, and Local Governments.
CO4: Relate constitutional provisions to technology, governance, and professional ethics.
Lecture Plan (14 Hours)

Module	**Topics**	**Hours**
1. Historical Perspective	Making of the Constitution, Constituent Assembly, Preamble	2
2. Salient Features	Federal structure, parliamentary democracy, secularism, rule of law	2
3. Fundamental Rights	Equality, Freedom, Constitutional Remedies	2
4. Directive Principles & Fundamental Duties	Purpose, examples, professional and civic relevance	2
5. Structure of Government	President, Prime Minister, Parliament, Governor, CM, Judiciary	2
6. Constitutional and Non-Constitutional Bodies	Election Commission, UPSC, Finance Commission, NITI Aayog	2
7. Constitution & Technology	Right to Privacy, IT Act, Cyber laws, governance in digital era	2

Suggested Activities (Non-contact hours / assignments):

- Case study discussion on landmark judgments (e.g., Kesavananda Bharati, Puttaswamy case).
- Reflection note on "Fundamental Duties and Engineers' Social Responsibility."
- Group discussion: "Balancing privacy and security in digital India."

Text Books:
Basu, D.D., *Introduction to the Constitution of India*, LexisNexis.
Pylee, M.V., *An Introduction to the Constitution of India*, Vikas Publishing.

References:
Bare Act – *The Constitution of India*, Ministry of Law & Justice.
PRS Legislative Research – https://prsindia.org
Supreme Court Landmark Judgments Database – https://main.sci.gov.in/